Software Requirements Specification

MIS

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1 Introduction

1.1 Document Purpose

Document Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for an AI-driven educational platform, focusing on assisting schools with inspection readiness. This SRS will serve as a blueprint for the development team and ensure alignment with stakeholder expectations. The primary objective is to outline functional and non-functional requirements necessary for the successful implementation of Use Case 1. This document builds upon discussions and insights from Workshop 1, referencing the attached workshop transcript for context. It will address system features, deployment models, stakeholder access, data handling, and hardware considerations.

The MSBC team shall use this SRS as a guide for developing the AI-driven school inspection readiness tool, ensuring that the developed solution meets the defined requirements and objectives. This document shall cover all four use cases initially, with Use Case 1 as the immediate focus for development. This SRS shall be supplemented by a Technical Design Document providing detailed infrastructure and platform specifications.

1.2 Project Purpose

Project Purpose

This Software Requirements Specification (SRS) details the requirements for an AI-driven education platform designed to support schools in various administrative and pedagogical functions. The primary goal of this project is to deliver an offline AI toolset that enhances efficiency and decision-making within top-tier international schools. The platform will offer modular, scalable solutions to address challenges such as scattered data, lack of integration, and inefficient workflows currently faced by educators and administrators. The system will generate data-driven internal recommendations, not public-facing reports. A key objective is to provide internal school documents to support internal improvement, not external publication.

The initial focus is on four key use cases: AI-driven school inspection readiness, curriculum enhancement and personalized lesson planning, academic performance tracking and visual reporting, and AI-enabled marketing and admissions support. A phased implementation approach is planned, beginning with core features and with considerations for potential expansion into other verticals, such as finance and human resources. The system’s success will be measured by its ability to centralize data, streamline processes, and demonstrably improve outcomes for schools.

1.3 Project Objectives

Project Objectives

The overall project objective is to develop modular AI tools for schools, focusing on inspection readiness, lesson planning, academic performance, and marketing support. These tools aim to address scattered data, lack of integration, and inefficient workflows. The system will initially prioritize offline deployment.

The following subpoints delineate the project objectives:

- AI-Driven Inspection Readiness: Support schools in preparing for inspections using a RAG system and automated analysis.
- Curriculum Enhancement: Assist teachers in creating resources and lesson plans, identify student gaps, and build personalized plans.
- Academic Performance Tracking: Develop tools to track and visualize academic performance, replacing current Power BI plugins.
- Marketing and Admissions Support: Enable AI-driven support for marketing and admissions, analyze school data to identify target segments.

The following pointwise requirements are defined for each subpoint:

- **AI-Driven Inspection Readiness**
- The system shall provide an interface for uploading inspection requirement changes.
- The system shall integrate previous school reports for analysis.
- The system shall allow users to ask questions related to inspection readiness.
- The system shall generate checklists and action lists.
- The system shall align PDF reports with KHDA formatting on demand.
- The system shall provide a chatbot for interactive follow-up queries.
- The system shall visualize insights like subject performance with charts and graphs.
- The system shall suggest actions based on inspection frameworks.
- The system shall generate internal recommendations, not public-facing reports.

- **Curriculum Enhancement**
- The system shall assist teachers in creating resources and planning lessons.
- The system shall aid in identifying student performance gaps.
- The system shall offer a knowledge base for curriculum-related queries.
- The system shall support personalized lesson plans based on student data.
- The system shall integrate with school management systems to pull data.
- The system shall support the integration of student attendance data.

- **Academic Performance Tracking**
- The system shall track attendance across schools and year groups.
- The system shall segment performance data.
- The system shall provide an interface for users to ask questions related to generated outputs.
- The system shall have a chatbot with the ability to answer follow-up questions based on a response.
- The system shall allow users to regenerate the report if it does not meet user expectations.
- The system shall generate a report identifying weakness of the data.

- **Marketing and Admissions Support**
- The system shall analyze academic performance data.
- The system shall analyze demographics from student management systems.
- The system shall analyze the conversion rates from marketing efforts.
- The system shall generate text-based suggestions for outbound materials.
- The system shall offer suggestions, text material at least, to schools for promotion.
- The system shall help schools with making appropriate marketing decision.
- The system shall allow schools to upload their marketing materials.
- The system shall provide charts and graphs to analyze their marketing campaigns.
- The system shall produce an output listing demographic data for school marketing.
- The system shall produce, at a minimum, the student demographics, which they can use to build campaigns.

1.4 Project Stakeholders

Project Stakeholders

Project Stakeholders	**Their respective Designation**
Riza Kaya	Founder/Client, Offline AI EdTech Visionary
Suhail Ahmed	AI & Technical Strategy Lead, MSBC Group
Ajay Chakrobarty	Business Analyst, MSBC Group
Shakti Gohil	Technical Consultant, MSBC Group
Sahil Sharma	Backend & AI Integrations, MSBC Group
Aesha Thakkar	Senior UI/UX Designer, MSBC
Ishika Anandani	UI/UX Designer, MSBC
Courage	*Designation Not Available*
Vedant	*Designation Not Available*
Shubhi Mittal	Meeting Organizer

1.5 Assumptions

Assumptions

The following assumptions underpin the software requirements specification for the AI-Driven School Inspection Readiness application:

- The primary deployment model is an offline device with a Linux-based system.
- The system's initial user base comprises school leadership and administration staff, approximately 5-8 individuals, who will have uniform data access privileges.
- The system will ingest data from multiple sources, including KHDA/ADEC framework documents, prior inspection reports, and MIS/LMS integrations for student data (attendance, grades, etc.).
- Schools will upload pertinent documents into local folders, with the possibility of future integrations with SharePoint, OneDrive, or APIs.
- The system outputs will include on-demand PDF reports in KHDA formatting, an interactive chatbot for document queries, data visualization tools (charts, graphs), and action items derived from inspection frameworks.
- The system shall function as an internal advisory tool providing data-driven recommendations rather than generating external public-facing reports.
- Onboarding processes must account for the variability in school systems and curricula by mapping school-specific data to standardized formats.
- Hardware optimization is necessary for AI performance, directly impacting responsiveness and tiered licensing models.
- Offline device updates will be managed either through physical file transfer or a local server connected to the cloud on a defined schedule.
- The hardware will be provisioned by the project team and will be tiered to accommodate varying user concurrency and data volume requirements.
- UI/UX design will initially focus on a chat-based interface with admin views for role-based access, file control, and user tracking.
- Data backup and recovery will be the responsibility of the school IT teams.
- API integrations with external systems like SharePoint and OneDrive are considered out of scope for the initial release but remain on the roadmap.
- The system will perform regular offline checks to ensure data integrity and availability, flagging missing data points and alerting relevant personnel.
- Users will be able to engage with the system using natural language to query uploaded documents and generate reports, supporting iterative refinement of outputs.
- A one-time onboarding wizard will guide users in setting up folder structures and identifying required data for inspection assessments.

- Base hardware specifications will support single-school usage, with customization options for storage and GPU power based on user concurrency and data volume.
- Data upload limits will be enforced to maintain system stability and performance, with tiered subscription models offering increased capacity.
- Yearly framework updates from regulatory bodies such as KHDA and ADEC are subject to change, with the system expected to accommodate document variations and updates.
- The system's scope is defined to encompass internal data and recommendations, excluding external research or consultations beyond the school's ecosystem.
- Initial integration of AI into the tool will focus on use case 1 only.
- The marketing tool, while offline, depends on the context prepared in previous use cases to function correctly and to provide an accurate recommendation to schools.
- Initial data will focus on generating student's information on reports.
- Reports will be in PDF format and include, data, graphs, and demographic data.
- Initial analysis will focus on enrollment, churn rate, and conversion.
- Data backup options will be passed back to the school's IT team.
- Limited uploading of 75-100 MB per document and a total upload limit per account.

1.6 Acronyms and Abbreviations

Acronyms and Abbreviations

Acronym	Abbreviation
ADEC	Abu Dhabi Department of Education and Knowledge
AI	Artificial Intelligence
AMC	Annual Maintenance Contract
API	Application Programming Interface
AWS	Amazon Web Services
BBC	British Broadcasting Corporation
CBSC	Central Board of Secondary Education
CRM	Customer Relationship Management
CSS	Cascading Style Sheets
DXV	Dubai Time
EAL	English as an Additional Language
EPS	Encapsulated PostScript
ERP	Enterprise Resource Planning
FA	Formative Assessment
GDPR	General Data Protection Regulation
GEMS	Global Education Management Systems
GPU	Graphics Processing Unit
HR	Human Resources
HTML	HyperText Markup Language
HOD	Head of Department
IAI	Internal Advisory Tool
IB	International Baccalaureate
ID	Identifier
IEEE	Institute of Electrical and Electronics Engineers

IFC	International Financial Centre
iSAMS	International School Assessment Management System
IT	Information Technology
KHDA	Knowledge and Human Development Authority
KPI	Key Performance Indicator
LMS	Learning Management System
LLM	Large Language Model
MIS	Management Information System
MOM	Minutes of Meeting
MSBC	MIS Business Consulting Group
MVP	Minimum Viable Product
MySQL	My Structured Query Language
NVIDIA	NVidia Corporation
NPU	Neural Processing Unit
PDF	Portable Document Format
PHP	Personal Home Page
Power BI	Power Business Intelligence
RAG	Retrieval-Augmented Generation
RAM	Random Access Memory
RDP	Remote Desktop Protocol
RTS	Relational Time Series
SAAS	Software as a Service
SAT	Scholastic Aptitude Test
SEN	Special Education Needs
SQL	Structured Query Language
SRS	Software Requirements Specification
STEM	Science, Technology, Engineering, and Mathematics
TPU	Tensor Processing Unit
UAE	United Arab Emirates
UI	User Interface
UML	Unified Modeling Language
USB	Universal Serial Bus
UX	User Experience
VPN	Virtual Private Network
VRAM	Video Random Access Memory

2 Non-Functional Requirements

2.1 Non-Functional Requirements

Non-Functional Requirements

- The system shall be designed to prioritize data security and user privacy, adhering to GDPR and local data protection laws (e.g., UAE, Dubai).

- The system shall support a minimum of 5 concurrent users on the base hardware configuration. Tiered subscription models shall offer support for 5-20 users and 20-50 users on higher hardware specifications.

- The system shall operate primarily on an offline, Linux-based device to ensure data security and privacy. A multi-tenancy, web-based deployment option shall also be considered.

- The system shall support role-based access control, with distinct privileges for system administrators, school leaders, teachers, and (where applicable) students.

- The system shall provide a user interface (UI) that is intuitive and user-friendly, inspired by the design of ChatGPT and other modern chat-based applications.

- The system shall have a defined upper limit to the amount of data any individual file upload can consume. 150MB is initially selected as the highest maximum upload size per file.

- The system shall have a maximum storage defined per school, with tiered subscription models which will allow the school to purchase additional storage.

- The system shall generate reports in PDF format, aligned with the formatting conventions of KHDA and ADEC inspection reports.

- The system shall generate responses to user queries in English, with Arabic translation as a future enhancement. The system shall offer options for male and female voices.

- The system shall include an admin dashboard providing file control, user tracking, and role-based access management.

- The system shall enforce data retention policies, automatically deleting or archiving data after a defined period following subscription termination.

- The system shall provide audit logs of user interactions and system events.

- The system shall support a single sign-on functionality via client app secret ID.

- The system shall compress all PDFs uploaded into the system.

- The system shall ensure that AI processing speed depends on optimized local hardware and that this will affect licensing tiers.

- The system shall flag gaps in required documentation and alert on the dashboard when there is missing data.

- The system shall have the ability to answer follow-up questions through the chat bot.

- The system shall have mandatory scheduled maintenance when framework changes.

- The system shall have a file upload limit.

- The system's chunking size shall be spot on.

3 User Personas

3.1 User Personas

User Personas

This section outlines the key user personas interacting with the AI-driven school inspection readiness system, detailing their roles, responsibilities, and expected interactions with the system. Understanding these personas is crucial for designing a user-centered system that meets the needs of all stakeholders.

- **School Leadership:** This persona represents senior leaders within the school, such as the principal and vice-principals.
- Responsibilities: Overseeing the school's inspection readiness efforts, identifying areas for improvement, and ensuring compliance with regulatory frameworks.
- System Interactions: Accessing comprehensive reports generated by the system, utilizing the interactive chatbot for specific inquiries, and using the system to make data-driven decisions. They shall be able to upload relevant documents and access the history of previous analyses.
- **School Administration:** This persona encompasses administrative staff responsible for day-to-day operations, data management, and coordination of inspection-related activities.
- Responsibilities: Uploading relevant documents to the system, managing user access, coordinating data integration efforts, and generating reports for school leadership.
- System Interactions: Uploading inspection documents (KHDA/ADEC framework, past reports), managing user profiles, setting up folder structures for data input, and

monitoring the system's performance. They shall have access to administrative dashboards to track user activity and manage system resources.

- **Teachers/Heads of Department:** This persona includes teachers and department heads who are responsible for curriculum delivery, student assessment, and continuous improvement within their respective subject areas.

- Responsibilities: Using the system to analyze student performance data, identify areas of weakness, develop personalized lesson plans, and improve teaching strategies. They shall provide feedback to the school leadership on areas needing attention.

- System Interactions: Interacting with the chatbot to analyze student performance, accessing relevant reports, uploading subject-specific data, and utilizing the system to generate suggestions for improvement. This data shall be restricted to approved sources.

- **Students (Future):** This persona will be added in future implementations. It will include students who would interact under supervised setups with individual logins for subject learning.

- Responsibilities: Interacting with the AI to gain help in learning, or assess their proficiency.

- System Interactions: Ask questions about subjects, access learning modules, and study their performance.

- The system shall provide user access control to allow or restrict access to the system based on the user's role.

- The system shall use a chat-based UI.

- The system shall support multiple languages such as Arabic and English for the UI.

- The system should ensure a balance between function and security to safeguard user data.

- The system shall provide templates that have been vetted to ensure that teachers that are non-technical can easily use the functions.

4 Scope

4.1 Scope

Scope

This Software Requirements Specification (SRS) defines the scope of the AI-driven School Inspection Readiness system. This system will provide an internal advisory tool leveraging AI to analyze school data and inspection frameworks, generate recommendations, and assist in preparing for inspections. It is not intended to create public-facing reports but to provide data-driven insights for school leadership and administration. The system will initially focus on supporting schools in the UAE, specifically those following the KHDA and ADEC frameworks. Future expansion to other regions and frameworks is a potential roadmap item. The project will prioritize an offline deployment model with consideration for a multi-tenancy web-based option in later phases.

• The system shall support the upload of relevant school documents, including KHDA/ADEC framework documents, past inspection reports, and MIS/LMS data extracts.

• The system shall provide system outputs, including PDF reports aligned with KHDA formatting and an interactive chatbot for follow-up queries.

• The system shall generate suggestions and action items derived from inspection frameworks.

• The initial user base will consist of school leadership and administration staff, with access limited to approximately 5-8 individuals per school.

• The primary deployment model shall be an offline device running on a Linux-based system, with hardware modularity to support varying user concurrency levels.

• Hardware and Scalability aspects will be Tiered Subscription Models, customizable storage, and GPU power depending on usage/concurrency.

5 Sequence Diagram

5.1 Sequence Diagram

Sequence Diagram

This section outlines the sequence of interactions between the actors and the AI-Driven School Inspection Readiness system for Use Case 1, focusing on the primary workflow of inspection preparation. The diagram illustrates the steps from user login to report generation, highlighting data flow and system responses.

The sequence begins with the School User initiating a login attempt through the System Interface.

The System authenticates the user against the User Database.

Upon successful authentication, the System grants access to the Inspection Readiness module.

The School User uploads relevant documents, such as KHDA/ADEC framework documents and past inspection reports.

The System receives the uploaded documents and stores them in the Document Repository.

The System initiates document processing, including chunking, indexing, and vectorization.

The School User interacts with the Chatbot Interface to pose questions regarding inspection readiness.

The Chatbot Interface forwards the query to the AI Engine.

The AI Engine accesses the Document Repository, retrieves relevant information, and formulates a response based on the ingested documents and pre-defined prompts.

The AI Engine returns the response to the Chatbot Interface.

The Chatbot Interface displays the response to the School User.

The School User may request the generation of a PDF report.

The System compiles relevant information based on the data within the Document Repository.

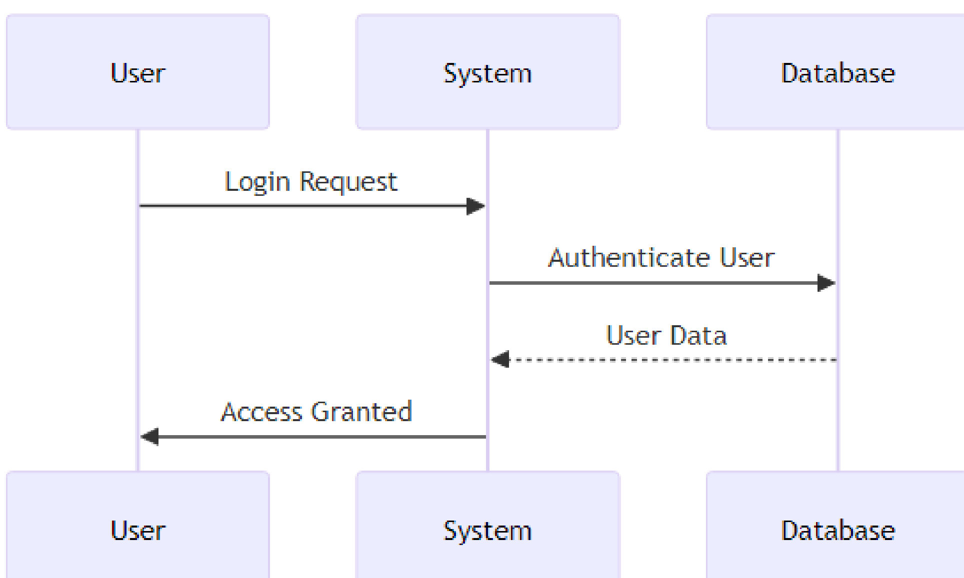
The System generates a PDF report aligned with KHDA formatting.

The System delivers the generated report to the School User through the System Interface.

The School User downloads the inspection readiness report.

The System stores the chat history and interactions within the Chat History Database for audit and personalization purposes.

The following diagram illustrates the system design:



6 Functional Requirements

6.1 Functional Requirements

Functional Requirements

- The system shall allow authorized school leadership and administrative staff to access the AI-driven school inspection readiness platform.
- The system shall provide a user profile and chat history for each user to personalize the experience.

* **The system shall ingest data from the following sources:**

- KHDA/ADEC framework documents
- Past inspection reports
- MIS/LMS integrations for student data, including attendance and grades
- The system shall allow schools to upload relevant documents to local folders.
- The system shall generate PDF reports on demand, aligned with KHDA formatting.
- The system shall provide an interactive chatbot for users to query uploaded documents.
- The system shall support visualization of insights, including charts and graphs for subject performance and trends.
- The system shall generate suggestions and action items derived from inspection frameworks.
- The system shall provide predefined prompts and suggested actions based on RAG-based analysis.
- The system shall maintain a history panel and a chat-based UI.
- The system shall support file uploads of PDF and image formats.
- The system shall implement data caching and vector-based search.
- The system shall include an admin dashboard with role-based access, file control, and user tracking.
- The system shall include an onboarding wizard for folder setup and required data list.
- The system shall operate primarily as an internal advisory tool providing data-driven recommendations.
- The system shall provide data mapping to standardized formats during onboarding to accommodate school variability.
- The system shall support both conversational interaction and downloadable inspection readiness reports.
- The system's performance shall be optimized based on local hardware specifications.
- The system shall cover Use Case 1 in full, including infrastructure and functional flows, in the SRS and Technical Design Document.

- The UI shall visualize chat, dashboard, and admin views.
- The system shall define required data fields for inspection assessment based on KHDA & ADEC handbooks.
- The system shall propose hardware spec tiers & costing models linked to user concurrency & GPU/RAM/storage needs.
- The system shall outline storage & caching strategy for offline data, especially for vector stores and chat histories.
- The system shall draft & deliver an SRS document covering infrastructure and functional flows for all identified use cases.
- The system shall document functional requirements, screens, visuals, validations, expected output, and workflow.
- The system shall cover technical requirements such as the operating system, dependencies, platform, services, and database.
- The server-side admin interface shall create and manage school instances and access information.
- The offline device shall synchronize with the server to update data.
- The system shall support file upload and email for device updates with an option for no external connections.
- The system shall provide a local server for offline application deployment connected to the cloud.
- The system shall provide a secure connection for remote device monitoring.
- The system shall enable scaling solutions focusing on single-school usage with the ability to enhance for multiple schools.
- The system shall allow leadership teams to upload documents and provide additional information for data evaluation.
- The system shall provide data evaluation based on predefined prompts.
- The system shall update frameworks yearly or tri-yearly and deal with document uploads despite the frequency of change.
- The system shall determine weaknesses in subjects and areas and function communicatively, similar to ChatGPT.
- The system's chatbot shall answer questions related to generated output.
- The chatbot shall have the ability to answer follow-up questions based on a response and remember past conversations.
- The system shall include a one-time onboarding wizard and a recommendation system based on handbook review.
- The system shall check the knowledge base for missing data points every midnight and alert all users of missing files via an automated email.
- The system shall allow schools to connect it to their SharePoint, OneDrive, or Google Drive accounts, if available.

- The system shall allow users to ask the system to regenerate content if it cannot produce it as per the user's expectations.
- The system shall have easy, predefined, and mostly used prompts.
- The system shall focus on weaknesses in previous reports, with an ability to change the sequence.
- The system shall include a dashboard that visually showcases the different flags, whether the school is performing in an aspect or not.
- The system shall allow the users to have their own profile where they can only see their details, not everyone else's.
- The system shall save the resources shared for use case one because they are all trying to get the same outcome.
- The system shall display performance outcomes against the school's framework.
- The system shall allow the user to ask the system to add an additional line item, such as a comparison chart, if the user has configured the format of the output.
- The system shall have a base template from where the user could use as is or change per their own will.
- The system shall recite the source when it answers the user back with the information.
- The system shall ensure that all add-on inputs be stored in a different vector store.
- The system shall include a top-down mandate that states, "This is what you need to do, this is what we're doing."
- The system shall be able to determine each school's outcomes and reports.
- The system shall be able to provide data-driven internal recommendations rather than public-facing reports.
- The system shall be able to support a chatbot and PDF reports.
- The system shall be able to function even while missing data points.
- The system shall be able to perform subject-targeted assessment.
- The system shall be able to recognize the differences between CBSC framework to the KHDA framework.
- The system shall have tier licensing that is dependent on local hardware.
- The system shall allow for limited storage and GPU power that depends on concurrency.
- The system shall have base hardware specification and expand from there based on the user amount.
- The system shall allow users to choose from easy predefined prompts or define their own prompts.
- The system shall provide a simple chat interface.
- The system shall include a super user side where the user can control who comes in, like an admin panel, and a panel for users using it, a history of what put in, and maybe a button

to speak, just to, so they don't want to type, they could speak, upload image, upload PDF, upload, you know, some of the different variables.

- The system shall store the image in a vector and can be temporarily or permanently stored.
- The system shall be able to accept an inspection report, either produced from the school itself or KHD/DEC one.
- The system shall be able to handle the upload of the curriculum and have the students' reports to verify it against.
- The system shall have an unbiased system to prevent personally uploaded documents to be handpicked and stabbed.
- The system shall have scheduled mandatory maintenance every time there is a framework change.
- The system shall have recommendation system, maybe a one-time onboarding wizard.
- The system shall have a folder structure within the device, where you drop in what is relevant to you.

7 System Requirements

7.1 User Interface

User Interface

The system shall provide a user interface that allows school leadership and administration staff (approximately 5-8 users) to interact with the AI-driven school inspection readiness platform.

The system shall present a chat-based interface, inspired by ChatGPT, for users to interact with the AI and query uploaded documents. This interface shall include a history panel to maintain context across interactions.

The system shall support file uploads of PDF and image formats, with file size limited to a maximum of 2MB per upload. The system shall allow for administrative control over user access, file management, and user tracking.

The user interface shall support role-based access control, displaying relevant features and data based on user permissions. The system shall include an onboarding wizard to guide users through folder setup and required data inputs.

The user interface shall initially focus on simplicity and ease of use, prioritizing functionality over complex visuals. Predefined prompts shall be provided to guide users and facilitate common queries.

7.2 Database

Database

The system shall utilize a database management system (DBMS) for persistent storage of data. The DBMS shall support offline operation on a Linux-based system to ensure functionality even without network connectivity.

The database shall be selected based on its compatibility with offline deployment, performance characteristics, and ability to handle structured and unstructured data. Possible options include MySQL, PostgreSQL, or a similar lightweight DBMS that aligns with system requirements.

The database shall store:

- User profiles and access permissions.
- Chat histories for personalized user experiences.
- School-specific data, including KHDA/ADEC framework documents, inspection reports, and MIS/LMS integrations.
- RAG-based analysis results, predefined prompts, and suggested actions.
- Cached files (PDFs, images) and vector-based search indices.

The system shall implement a storage and caching strategy optimized for offline data, focusing on efficient retrieval of vector stores and chat histories. This strategy shall account for limited storage capacity on local devices.

The system shall support customizable storage allocation and tiered subscription models based on data volume and user concurrency to accommodate varying school sizes and usage patterns.

The database shall employ encryption to protect sensitive data, particularly when stored on local offline devices.

7.3 Communication/Security

The system shall provide secure and reliable communication channels for data exchange between system components and authorized users. The system shall employ industry-standard encryption protocols to protect sensitive data during transmission and storage, both in offline and online deployments.

• The system shall restrict unauthorized access to system resources and data through role-based access controls, limiting privileges to only those necessary for each user's function. School leadership and administrators shall be granted access to all data relevant to their responsibilities. The system shall maintain a detailed audit log of all user interactions, including data access, modifications, and system configurations, to ensure accountability and facilitate security investigations.

• The system shall support offline operation on Linux-based devices, ensuring functionality and data security even without an active internet connection. Data synchronization mechanisms between offline devices and central servers shall be secure and configurable to accommodate varying school policies, including options for scheduled synchronization or physical file transfer. The system shall protect against data breaches by requiring that the offline device have encrypted storage.

7.4 System Integration

System Integration

The system shall support integration with various school data sources, including but not limited to PDF reports, Excel sheets, and APIs for MIS and LMS systems (e.g., iSAMS, Veracross). The system shall facilitate the extraction and transformation of data to ensure compatibility with its internal data model. The system shall provide a mechanism to schedule data synchronization from external sources, with configurable intervals based on school needs. The system shall support CSV upload for bulk student data ingestion, such as names, forms, email addresses, and dates of birth.

The system shall integrate its modules via a unified, role-based access control, where access to each module will be managed by school administrators. The system shall ensure its user interface dynamically adjusts based on a user's assigned roles and the school's active subscriptions. The system shall allow user access to multiple modules based on established configurations. The system shall implement separate authentication methods and encryption protocols for cloud-based and offline deployments to ensure data security.

The system's design shall support an admin interface for monitoring and managing system usage, including but not limited to the number of users, number of active subscriptions, and subscription expiration dates. The system shall maintain an audit log of chat interactions and document accesses for security and compliance purposes. The system shall provide customizable prompts for users, which are preconfigured and adaptable per user department or role.

7.5 Constraints and Limitation

Constraints and Limitations

• The system shall be designed primarily for offline deployment, with an initial focus on single-school usage to ensure data privacy and security.

- The system's AI processing capabilities shall be limited by the local hardware specifications. Hardware scalability will be customizable and tiered to support varying levels of user concurrency and data volume, which affects licensing tiers.

- The system shall be implemented using a modular architecture, enabling schools to select specific use cases based on their needs. Each module will function independently, providing flexibility but also limiting cross-module functionality if not explicitly included in the selected package.

- The system shall support integrations with major school management systems (MIS) and learning management systems (LMS) via APIs, or file imports, focusing on essential data fields such as attendance and grades. Data synchronization will be scheduled rather than live to maintain offline capabilities.

- The system shall provide support for publicly available ADEC and KHDA inspection frameworks, ensuring alignment with regulatory requirements. Custom frameworks and data sources may require additional configuration and development effort.

- The system shall prioritize predefined prompts and suggested actions based on RAG-based analysis to provide targeted recommendations, ensuring the AI functions as an internal advisory tool rather than generating public-facing reports.

- The system shall primarily focus on textual content, images, tables, and graphs, excluding video support in the initial phase.

- Individual user uploads shall be limited to a maximum file size of 150 MB to ensure system performance and efficient resource allocation.

- The system shall be limited in its capacity to handle external data sources beyond the school's internal data, such as comprehensive external marketing strategies, due to the focus on offline functionality and data privacy.

- The initial focus shall be on the English language, with planned support for standard Arabic through translations. Full natural language processing support for Arabic shall be explored in later development phases.

- The system's reporting capabilities shall be limited to PDF generation and interactive chatbot support, providing both conversational interaction and downloadable reports. The system will not offer full-scale Power BI capabilities initially.

- The system shall adhere to GDPR regulations, UAE data privacy laws, and Dubai's Internet use regulations. Compliance mechanisms and data retention policies must be implemented and clearly communicated to users.

7.6 Legal

Legal

The system shall comply with the General Data Protection Regulation (GDPR) to protect user data privacy. The system shall also adhere to data privacy and internet usage laws as mandated by the United Arab Emirates and the Dubai International Financial Centre. System design and operation shall incorporate mechanisms to meet these legal requirements.

- The system shall include an audit logging mechanism that monitors data changes, user access, and generated outputs for accountability and compliance.

8 System Architecture

8.1 Overview

Overview

The software requirements specification (SRS) will outline all system features, functionality, and design requirements for developing AI-driven education tools. These tools will support schools in areas such as inspection readiness, curriculum enhancement, performance tracking, and admissions. The primary focus is on creating modular, scalable solutions that address real school problems, including data centralization and workflow efficiency. The system shall initially focus on use cases relating to internal school processes, with potential expansion into external integrations.

The system shall offer both offline and cloud-based deployment options. The offline system shall prioritize data privacy and security, with updates managed through controlled mechanisms. The cloud-based system shall provide wider accessibility and potential for multi-tenancy. The system's architecture shall accommodate hardware and software specifications to meet the demands of varying school sizes and usage scenarios.

8.2 AI Subsystems

AI Subsystems

This section details the AI subsystems required for the AI-driven education platform, focusing on the functional and non-functional requirements necessary to support school inspection readiness, curriculum enhancement, personalized lesson planning, and admissions/marketing support. The AI subsystems will integrate with existing MIS and LMS systems to provide data-driven insights and recommendations.

The system shall include an AI-driven inspection readiness subsystem:

- This subsystem shall utilize a Retrieval-Augmented Generation (RAG) system to analyze inspection documents.
- The system shall enable the upload of previous school reports and inspection framework documents.
- The system shall support the generation of internal school reports aligned with KHDA/ADEC formatting.
- The system shall include an interactive chatbot interface.
- The chatbot shall allow users to ask questions about uploaded documents.
- The chatbot shall maintain chat history for personalized user experience.

The system shall include a curriculum enhancement and personalized lesson planning subsystem:

- This subsystem shall integrate with existing MIS systems to access student data (attendance, grades, etc.).
- The system shall support the creation of personalized lesson plans based on student performance.
- The system shall provide access to a resource database for teachers.
- The system shall support the identification of student weaknesses and strengths.
- The system shall support curriculum enhancement and personalized lesson planning using data from MIS systems.

The system shall include an AI-enabled marketing and admissions support subsystem:

- This subsystem shall support the analysis of student demographics and application data.
- The system shall provide recommendations for targeted marketing campaigns.
- The system shall support the analysis of competitor school marketing materials (via PDF upload).
- The system shall facilitate the preparation of questions for entrance exams.
- The system shall support the evaluation of results of entrance exams.

The AI system shall adhere to the following requirements:

- Shall function as an internal advisory tool, providing data-driven recommendations rather than public-facing reports.
- Shall be designed to handle unique school systems and curricula, requiring initial data mapping to standardized formats.
- Shall optimize performance based on local hardware specifications, with tiered licensing models reflecting processing power and data volume.
- The AI system shall support the processing of various file formats, including PDF, Word, and Excel.
- Shall cite the source for output of its analysis.

The system shall feature the ability to connect to external services:

- Shall offer SharePoint, OneDrive, and Google Drive API integrations for data input, with optional custom development.
- Shall have API integrations with major MIS providers (e.g., iSAMS, Veracross), though data will be pulled, not written back.

The system shall have a modular architecture, with each use case functioning independently, and:

- Shall require data centralization via a unified repository and API layer for efficiency.
- Shall provide a lean build with a roadmap for future customization through portals and configurations.

Regarding deployment, the system shall accommodate both offline and cloud-based models:

- Shall prioritize offline deployment on a Linux-based system for security and privacy.
- Shall support multi-tenancy web-based offerings with a subscription model as a secondary option.
- Shall have customizable hardware specifications based on user concurrency and data volume.

The system shall maintain the following non-functional requirements:

- Shall comply with GDPR, UAE data privacy laws, and Dubai regulations.
- Shall maintain data security through encryption and access controls.
- Shall provide role-based access control to system features and data.
- Shall support a history panel for each user to track their interactions.

Finally, regarding testing and user feedback, the system:

- Shall undergo beta testing with selected schools to refine document structure and response quality.
- Shall incorporate a feedback mechanism for users to rate the quality of AI responses.
- Shall allow for adjustments to the AI models based on feedback and performance data.

9 Hardware Requirements

9.1 Hardware Requirements

Hardware Requirements

- The system shall be designed to primarily operate on an offline device.
- The system shall support modular hardware upgrades to accommodate varying user concurrency levels and data volumes.
- The hardware shall be customizable in terms of storage and GPU power.
- The system shall operate on a Linux-based system to avoid licensing costs.
- The hardware solution shall support tiered subscription models based on data volume and user concurrency.
- The system shall support tiered hardware specifications based on concurrent users: 1-5, 5-20, and 20+.
- The system shall provide recommendations for hardware specifications to ensure optimal AI processing speed and responsiveness.
- The system shall be designed to function on a single-school, single-device configuration as a base unit.
- The hardware solution shall facilitate cloning for rapid deployment to multiple schools, whether via USB stick or other cloning methods.
- The system shall accommodate hardware changes based on school requirements, such as GPU or AI processing chip upgrades.
- The system shall provide hardware options in different sizes, ranging from small boxes to mid-sized servers and full server racks, to service varying school sizes and usage patterns.
- The hardware solution shall be optimized to minimize AI processing time, manage token throughput, and accommodate concurrent users.
- The system shall be compatible with neural processing units (NPUs) and tensile processing units (TPUs) for AI processing.
- The system shall provide a base hardware specification for 1-5 concurrent users.
- The system shall support modular upgrades, such as additional GPU VRAM and storage, at defined cost increments.

- The system shall offer a top-down mandate solution option that can be directly pushed by inspection government into the schools.
- The system must have an unbiased setup, and some unbiased solution that will prevent people to do bad things
- The system shall provide hardware solutions designed such that only AI processing will take place.
- The hardware should have mandatory maintenance scheduled for every school when a new framework change takes place.
- The system shall have default templates that would be base level, however, if they are any changes required there would be a one week turn around period for the change request.
- The system should ensure data privacy via on-premise hardware, and provide recommendations on how the IT team in the school can configure the device.
- The system shall have offline devices, and an API, and have the ability to send data over, while still keeping data secure.
- The system's hardware setup should be designed in a way to accelerate the speed of the system.
- The system's performance should be benchmarked to be on par with a system as powerful as Cloud or ChatGBT.
- The system should take in consideration that the computer infrastructure of a standard school would not be a sophisticated as some of these foundational model providers
- The system hardware should be optimized for offline usage so that the customer can have access to the product even without a wifi connection
- The hardware should account for local networks.
- The hardware needs to account for different data size quotas.
- The hardware should have the ability to extract chips and put them together so that chips are only processing for AI.
- The server has to be able to handle admin when school is created.
- The server must be able to handle the ability for an admin to manage access for the client.
- The server and software must be able to figure out how updating the device will take place
- The server needs to figure out data storage and caching strategies for offline data, especially for vector stores.
- The device should also be able to connect to the schools network
- The hardware must give the customer the ability to connect to the device and make sure it is a secure connection
- The hardware must have a secure mechanism where the device can clone the configuration for multiple schools
- The system needs to factor in storage options and cost for all data.
- The data on each device will need to be secured and should be encrypted.

• The system requirements regarding GPU and other dependencies need to be documented well.

10 Acceptance Criteria

10.1 Acceptance Criteria

Acceptance Criteria

The AI-Driven School Inspection Readiness platform shall generate PDF reports aligned with KHDA formatting requirements upon user request. The generated reports shall determine weaknesses in each subject and area, enabling users to identify areas for improvement.

• The system shall include an interactive chatbot interface, similar to ChatGPT, that allows users to ask follow-up questions related to the generated output. The chatbot shall retain the context of the previous conversation to enable users to chain questions together. The system shall display weaknesses in previous reports as a priority, highlighting areas that need improvement.

• The hardware specifications shall be tiered based on user concurrency, with base specifications for single-school usage and customizable storage and GPU power based on usage and concurrency requirements. The system shall support a modular approach to future customization options, including SharePoint, OneDrive, or API integrations.

11 Product Future

11.1 Product Future

Product Future

The system's future development shall focus on expanding its capabilities and market reach while maintaining a modular and adaptable architecture. Future iterations shall consider integration with a wider array of systems, including finance and HR, to provide comprehensive solutions for schools.

• Future product iterations shall include enhanced accessibility features, catering to a broader range of user needs.

• The system shall be expanded into diverse geographical regions, with localized content and language support.

• Top-down mandates from educational boards to integrate with school systems are a long-term goal.

• To reach new markets, future versions shall explore new hardware possibilities, for example, camera integration for visual analysis of UAE school cleanliness.

• A financial analysis function shall be developed to evaluate marketing campaign revenue and expenses.

12 Open Queries

12.1 Open Queries

- What is the preferred method for schools to manage media files, considering storage capacity limitations? Should the system provide recommendations for local storage solutions?
- Given the emphasis on offline functionality, how will the system handle updates to AI models, data sources, and system logic? What is the update mechanism for devices that might not have regular internet connections?
- To ensure proper security and ethical use of the system, what specific data governance policies (data retention, data access) will be enforced, especially concerning personally identifiable information (PII) of students and staff?
- For the proposed multi-tenancy web-based offering, what is the strategy for capacity planning and resource allocation to ensure consistent performance across different subscription tiers?
- Concerning system outputs, is it possible to integrate additional output formats, such as Word (.docx) files, to match current school reporting standards?
- Regarding the file upload feature (PDF, image), what is the maximum file size to prevent performance issues? Is there an automated compression mechanism that can be implemented before vectorization?
- Given the potential hardware performance limitations and varied network setups across schools, what specific steps can be taken to optimize the system to provide the required performance in various school environments?
- What is the anticipated process for data backup on the offline device, and what role will the school's IT team play in securing the system within their internal network?
- How do schools currently handle updates to inspection frameworks from authorities like KHDA and ADEC? Does the system need to support a manual upload process for these documents, or can we explore automated API integration?
- How variable is the content based on the user role (e.g., senior leadership vs. subject head)? Is there a matrix of data and content accessibility by different user roles?
- What mechanism (e.g., folder structure, naming conventions) will ensure that internal school documents, used to prepare for KHDA inspections, are accurately mapped to the standardized reporting framework?

- Does the system flag missing data points required for a full KHDA/ADEC inspection assessment, and should the system notify relevant stakeholders via dashboard or email?
- Can the user ask follow-up questions to the chatbot and generate customized PDF reports?
- What should the recommendation system or onboarding wizard include to ensure a smooth initial setup for each new school?
- How will the system facilitate the extraction of student-related data from diverse school management systems, given the unique structures and potential access challenges (API integration, etc.)?
- How frequently should system-generated reports be produced and how do we incorporate periodic update?
- What hardware specification tiers should be proposed, covering variations in user concurrency, data volume, and AI processing requirements?
- What process is to be used to collect internal school documents (Excel/PDF/Word) from four to five schools, and will these be representative of data and operational practices?
- Given the importance of the RAG system, what chunking size and continuity strategies should be employed? How do we handle the processing of images contained in uploaded documents?
- While considering UI/UX, is there a standard size that the user must be restricted to when they upload a document?
- Can the system flag those PDF files that the system has previously generated verses those that have been uploaded?
- How is the system going to verify uploaded curriculum given any handpicked student reports?
- What level of detail is required in the SRS to guide software developers, considering the evolving nature of the project and reliance on MSBC for design and implementation advice?
- Can the team clarify whether this marketing module will support offline environments?
- Are text-based suggestions with analytic graphs preferred, as opposed to heavy media generation?
- Is integration with prior use cases vital to support marketing decisions?
- Should we add finance as another area that we can phase into?
- What is the current preferred UI/UX? Chat or portals?
- Can team members discuss what it would take to do basic churn and enrollment analysis?
- Do the schools being consulted have an existing data management system?
- What are each of the data points that would be required for each unit?

- Would the content change depending on who is accessing it and will everyone see the same thing?
- How often do the frameworks change, if different boards have different frameworks and does the system need to keep track of those as well?
- Will there be a specific section or sections that need to be most important, like weaknesses, in the generated output?
- Will the system have a security framework that will allow a system admin to have group access, where each individual has a profile to chain conversation tracking?
- Define the data points for each individual unit clearly.
- If schools do not care about the correlation, such as maybe between economics and marketing, would they not upload the data?
- What level of security is to be given to offline data from a potential on-site device?
- If share point or a one drive integration is implemented, how do we sell this solution?
- How can we determine if any additional documents, along with additional information, is provided to ensure the system can evaluate the data?
- Should the system determine weaknesses in each subject and in each area so that the system should be as communicative as the cloud or chat GPT?
- Can the system produce follow-up questions, based on a response, and have the ability to regenerate if it is unable to produce as per the user's expectations?
- How can we restrict the user to upload the document at a certain size, and should the size of the document directly impact the system?
- Implement SRS document covering all use cases.
- Prepare UI/UX design concepts based on standard chat interface.
- Prepare and share policy/workflow documents for integration.
- Define CSV/PDF upload capabilities and usage for analytics.
- Prepare estimated cost and project timeline post SRS approval.
- The product shall have a goal of offline AI tools to support schools in inspection readiness, lesson planning, personalized teaching, and admissions/marketing insights.
- The product shall target top-tier international schools with a lean focus on modular, scalable solutions.
- The AI tools shall support AI-driven inspection readiness support, curriculum enhancement, personalized lesson planning, academic performance tracking and AI-enabled marketing and admissions support.
- The data shall be sourced from PDF reports, MIS, and LMS systems.
- API integrations with major MIS providers (e.g., iSAMS, Veracross) shall be available.
- Offline deployment preferred, with optional cloud-based solutions for wider adoption.

- Dynamic, customizable reports (e.g., PDF/Word formats) shall be generated.
- Each use case shall function as an independent module to allow flexibility for schools.
- A unified repository and API layer are essential for efficiency and user experience.
- There shall be a Lean build with a roadmap for future customization options through portals/configs.
- There shall be rapid MVP development in response to high demand from schools.
- The platform shall address scattered data, lack of integration, and inefficient workflows.
- The system shall support text, images, tables, and graphs, and video capability to be considered for the future.
- There shall be a limitation of 10GB storage per subject and 2GB per teacher.
- PDFs may undergo compression during ingestion.
- The system shall emphasize GDPR and regional compliance.
- There shall be a central repository accessible by all users, with personal memory per teacher, with an option to share curated material.
- Offline systems shall have manual or semi-automated licensing.
- Expiry policies shall include grace periods and automated data deletion.
- The system shall have licensing and subscription management.
- There shall be access controls for various user types.
- The system shall provide generation only, not storage/distribution.
- The primary interface shall be in English, with Arabic support through translations.
- The system shall allow for 5-10 concurrent users per subscription level and scalable hardware options.
- The system shall have UI that adapts according to permissions and active subscriptions.
- The platform shall have new unified design for multiple use cases.
- The platform shall integrate configurable modes and document generation.
- The system shall limit each user to access, governed by teacher/admin assignment and restricted to classroom environments, to ensure safety.
- The system shall provide different hardware based on use and the schools' influence and need.
- The system shall allow students to interact under supervised setups with individual logins; future possibilities include VPN access.
- The system shall allow optional inclusion of formative assessments.
- The system shall allow tagging and categorization will help contextualize learning materials.

- The student's information or data shall be deleted from the system upon request.
- Uploads shall be on-demand, but processing may be scheduled to optimize compute resources.
- The system shall integrate basic voice interactivity.
- The system shall integrate visual elements like mind maps.
- There shall be strict rules against parent access or home usage in initial phases and future flexibility with multi-tenant models.
- Teachers shall define interaction tones and teaching styles, supported by pre-configured templates.
- The system shall support feedback on responses for fine-tuning.
- The system shall implement custom prompts with moderation of content and student safeguards.
- KPIs and analytics shall be simple and expandable.
- LLM safeguards and teacher-curated data sources shall restrict AI responses and maintain alignment with curriculum.